

Codeforces and Polygon may be unavailable from [February 22, 6:30 \(UTC\)](#) to [February 22, 20:30 \(UTC\)](#) due to technical maintenance.

×

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

C. Line

time limit per test: 1 second
memory limit per test: 256 megabytes

A line on the plane is described by an equation $Ax + By + C = 0$. You are to find any point on this line, whose coordinates are integer numbers from $-5 \cdot 10^{18}$ to $5 \cdot 10^{18}$ inclusive, or to find out that such points do not exist.

Input

The first line contains three integers A, B and C ($-2 \cdot 10^9 \leq A, B, C \leq 2 \cdot 10^9$) — corresponding coefficients of the line equation. It is guaranteed that $A^2 + B^2 > 0$.

Output

If the required point exists, output its coordinates, otherwise output -1 .

Examples

input	Copy
2 5 3	
output	Copy
6 -3	

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 7

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms) ▼

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
362789420	Feb/13/2026 18:37	Accepted

→ Problem tags

math number theory *1800

No tag edit access

→ Contest materials

- Codeforces Beta Round #7